Survivability

Continuity of Intelligence Under Infrastructure Failure

The mathematical foundation for the Survivability Engine: the degradation lattice L0–L4 as a chain of capability envelopes, descent and ascent predicates, the safety-equivalence invariant, and survivability as a design discipline rather than a hope. Descent is deterministic; re-ascent is gated by an integrity validator.

Dr. Christabel Odeta — SAVANT / CLAI-OS / AEGIS-GLOBAL
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C0 INFORMATIONAL · MATHEMATICAL SPECIFICATION — ALL PROPERTIES STATED AS PROOF OBLIGATIONS UNTIL DISCHARGED

DOCUMENT CONTROL & EPISTEMIC CONTRACT

0.1 Provenance and Position in the Program

This is Paper 7 of the research-and-build program defined by the *SVT Architect Analysis*: the survivability paper. The source's contribution is a single distinction, and it is the one this paper exists to formalize: *graceful degradation means losing capability without losing constitutional safety* — a much stronger engineering claim than "offline AI," because it constrains not what the system can do when the infrastructure fails but what the system *may* do, at every level of failure. The degradation lattice L_0–L_4 (cloud, local, edge, paper, human) appeared in Paper 0 as a list; here it becomes a chain of capability envelopes with descent predicates, recovery semantics, and a measured invariant. It reconciles against all seven predecessors under the standing rule: stricter constraint wins, adoptions recorded (§11).

0.2 Epistemic Contract

Unchanged: VERIFIED-FROM-SOURCE, FORMALIZED-HEREIN, CONJECTURE, UNKNOWN. All named properties are proof obligations. One sharpening for this paper's subject: survivability claims are the easiest to assert and the hardest to test, because "the system degrades gracefully" is invisible until the day the power goes out. Every survivability claim here is therefore tied to a descent predicate, a deferred-obligation ledger, or a recovery log entry — an artifact that exists before the failure, not a story told after it.

0.3 Authority Boundary

C0 informational. This document specifies how authority and capability contract under infrastructure failure; it authorizes no fallback protocol, defines no clinical contingency, and certifies no deployment. The paper tier (L_3) specified here is a mathematical object — an envelope with contents and staleness semantics — not a printable clinical document. Certification of any actual paper artifact belongs to the governance of the deploying institution.

0.4 The Thesis in One Sentence

Every prior paper asked what the system must be while the infrastructure holds; this paper asks what must still be true when the infrastructure does not — and answers with an invariant: the safety properties descend intact through every level of collapse, because they were never properties of the infrastructure, only of the constitution.

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PART I — THE SURVIVABILITY PROBLEM

1 Failure Is a Design Condition, Not an Accident

The conventional architecture treats infrastructure failure as an exception: an alarm, an error page, a status dot gone red. The deployment domain of this program treats it as a certainty: networks partition, power fails, devices are stolen, clinics run on generators, and the patient in front of the system does not pause while any of it is repaired. Paper 3 gave absence its semantics; this paper gives the largest absence of all — the absence of the infrastructure itself — its calculus. FORMALIZED-HEREIN

1.1 Why “offline AI” is the wrong claim

The industry's survivability claim is capability-shaped: *the model still runs offline*. It is a claim about what the system can do. The source's claim is constitution-shaped: *the safety properties still hold offline* — a claim about what the system may do, and the two are not only different but opposed in their engineering consequences VERIFIED-FROM-SOURCE. A system optimized for the first claim carries its full behavioral repertoire into degraded conditions and discovers, mid-failure, which behaviors depended on infrastructure it no longer has: the guideline check that needed the network, the audit that needed the cloud log, the verification that needed the verifier. A system built for the second claim has already answered those questions at design time: every behavior is tagged with the level at which it remains lawful, and the answer survives the failure because it preceded it.

1.2 Inherited machinery

Nearly every mechanism needed is already built; the paper's work is to aim them at infrastructure. From Paper 3: the A-2 absence class (network, power, compute) with its detector obligations — the lattice's descent triggers are absence detections, and a level change is an absence with an authority consequence. From Paper 1: the degradation levels L0–L4 as lifecycle states — now promoted from list to lattice. From Paper 5: evidence decay — every cached artifact in a degraded tier is evidence with a clock running. From Paper 6: interim mode — the objective engine's behavior when certificates cannot be refreshed is the objective face of infrastructure failure, and §6.2 composes the two. From Paper 4: the audit stack must itself survive — §4.4 specifies which audit depth each level can afford. The survivability engine is the constitution, compiled for the day the lights go out FORMALIZED-HEREIN.

1.3 Design decision D-8.1: failure is a state, not an event

> **D-8.1 — DEGRADATION IS CONSTITUTIONAL STATE**
>
> *The degradation level is part of the machine's operative state — held in the same registry style as authority, assumptions, evidence, and objectives; visible to every gate evaluation; and governing which behaviors are lawful before any of them execute. It is not an operational flag consulted by exception handlers. It is a constitutional quantity with detectors, transitions, and invariants.*
>
> FORMALIZED-HEREIN

The consequence is architectural: nothing “checks whether the network is down” at the moment it needs the network. The gate already knows the level, because the level is part of the state the gate evaluates, and a behavior whose prerequisites exceed the level never reaches the attempt.

1.4 The two clocks of failure

Infrastructure failure runs on two clocks, and the architecture keeps them distinct because they carry different dangers. The *descent clock* is fast: a partition is detected in seconds, and the level transition is immediate — latency here is exposure. The *ascent clock* is slow: restoration is not the mirror of loss, because the world kept moving while the system was degraded — evidence aged, deferred obligations accumulated, and the state the system returns to is not the state it left (§5, §7). Architectures that confuse the clocks fail in both directions: slow descent bleeds authority into conditions that cannot support it; fast ascent pretends the degraded interval never happened. Both clocks get their own predicates, and neither borrows the other's urgency FORMALIZED-HEREIN.

PART I — THE SURVIVABILITY PROBLEM

2 The Degradation Lattice

2.1 The five levels

The source fixes the chain VERIFIED-FROM-SOURCE:

$$L_0\,\text{Cloud} \;\to\; L_1\,\text{Local} \;\to\; L_2\,\text{Edge} \;\to\; L_3\,\text{Paper} \;\to\; L_4\,\text{Human} \tag{2-1}$$

L_0 Cloud: full infrastructure — network, compute, verification services, the complete registry with live refresh. **L_1 Local:** the institution's own systems hold; external network is gone — local registry, local verification, no cloud audit channel. **L_2 Edge:** the institution's infrastructure is gone; the device itself is all that remains — battery or solar budget, cached registry, on-device detection only. **L_3 Paper:** no computation — the system's contribution is the artifact it emitted in advance: the paper tier, a physical document whose contents were lawful at emission and whose staleness is printed on its face. **L_4 Human:** the system's contribution is what it transferred before the collapse — the trained practice, the cached judgment aids, the handoff record; authority has fully returned to human institution. Each level is a *capability envelope*, not a device class VERIFIED-FROM-SOURCE: the same hardware can sit at different levels, and the envelope — not the box — determines what is lawful.

2.2 The level object

> **L-OBJ — LEVEL OBJECT**
>
> *Each level L_k is the record $\langle C_k, \Sigma_k, \mathcal{I}_k, \tau_k^{\text{cache}}, \text{pred}_k\rangle$: the capability envelope C_k (the set of behaviors lawful at this level); the safety properties Σ_k that must hold; the infrastructure dependencies $\mathcal{I}_k$ whose presence defines the level; the maximum cache age τ_k^{cache} beyond which cached artifacts stop counting as support; and the descent predicate pred_k whose satisfaction places the system at this level. The level is certified by detection, not by declaration: the system is at L_k when pred_k holds and pred_{k-1} does not.*
>
> FORMALIZED-HEREIN

2.3 Capability contracts, safety holds

The chain orders capability strictly and safety by equivalence — the source's invariant, with its quantifier made explicit in §3:

$$C_0 \supsetneq C_1 \supsetneq C_2 \supsetneq C_3 \supsetneq C_4 \qquad \text{and} \qquad \Sigma_0 = \Sigma_1 = \cdots = \Sigma_4 \tag{2-2}$$

The strict inclusions are the honest part: each level genuinely loses capability, and the architecture says so — L_3 cannot compute, L_4 cannot even emit. The equality of the Σ_k is the

radical part, and §3 gives it the only meaning that can survive inspection: the same safety properties, evaluated against the level's own capability envelope.

2.4 Why envelopes, not switches

The naive implementation of degradation is a set of feature flags: offline mode turns off the network-dependent features. It fails in the way flags always fail — combinatorially and silently, because the flags encode pairwise dependencies while the failures arrive as wholes. The envelope inverts the representation: each level names what is *lawful*, and everything unlisted is unlawful at that level by default. The difference is the negative-space move of Paper 3 applied to capability: enumerate the permitted, not the forbidden, because the permitted set shrinks under failure while the forbidden set explodes — and a system that must enumerate what it may no longer do, mid-collapse, will miss one FORMALIZED-HEREIN.

PART II — THE INVARIANT

3 The Safety-Equivalence Invariant

The source writes the invariant with an approximation sign, and the honesty of that sign matters: $\text{Safety}_{L0} \approx \text{Safety}_{L1} \approx \ldots \approx \text{Safety}_{L4}$. This section gives the approximation its semantics — what is equal, what is measured, and what happens when the measurement fails. VERIFIED-FROM-SOURCE in form; semantics FORMALIZED-HEREIN

3.1 What "equal" cannot mean

Three readings of the invariant must be rejected at the outset. Not *equal outcomes*: L_4 cannot produce what L_0 produces, and pretending otherwise is the capability-shaped lie again. Not *equal guarantees*: the properties are enforced by different machinery at each level, with different strength — the cloud verifies continuously; the paper tier was verified once, at emission. Not *equal effort*: holding the properties at L_2 on a solar budget is harder than holding them at L_0, and the architecture prices the difficulty rather than hiding it. What remains is the only reading worth having FORMALIZED-HEREIN.

3.2 The invariant, with its quantifier

> **I-SE — SAFETY EQUIVALENCE ACROSS LEVELS**
>
> *I-SE's statement is restated in Appendix H-F (p. 32): the quantifier runs over system-influenced decisions, with occupancy floors making the invariant non-vacuous at every level.*
>
> FORMALIZED-HEREIN — invariant in the constitutive reading, measured form. CONJECTURE pending B-33 (H50). Killed by: a deployment trace in which an action lawful at $L_k > 0$ violated a σ that held at L_0 beyond tolerance.

The italicized clause is the spine. An action may be unavailable at L_3 because L_3 cannot compute it — capability. It may never be available at L_3 while being unsafe at L_0 — because then the level would be laundering risk through degradation, and the failure of the infrastructure would have become a loosening of the constitution. The lattice only descends; it never loosens.

3.3 Safety-critical continuity

Some behaviors are not merely lawful at low levels but *required* — the safety-critical continuity set: refusal when the evidence cannot support action, the null output on custody failure, the

transfer instinct made physical (a printed escalation path at L_3), the recording of what was done for later reconciliation. Continuity obligations are the floor of each envelope, and they have a distinguishing feature: they must be executable with zero infrastructure, because at L_3–L_4 zero infrastructure is the condition. A continuity obligation that silently required the network would be a contradiction the failure would eventually find FORMALIZED-HEREIN.

3.4 The measurement obligation

> **S-1 — EQUIVALENCE IS MEASURED, NOT PRESUMED**
>
> *Each level's violation rates per property are tracked in the deployment log against the L_0 baseline; a level whose measured v_σ exceeds tolerance is placed in restricted standing — its envelope shrinks to the subset whose safety is measurably intact — pending redesign. The invariant is therefore not a claim the architecture makes about itself but a claim the deployment log continuously tests.*
>
> FORMALIZED-HEREIN — obligation; the survivability analog of Paper 5's calibration loop, pointed at levels instead of parameters.

S-1 is what separates the invariant from a slogan. Every safety architecture asserts that its degraded modes are safe; this one states the assertion as a measurable relation, assigns it a tolerance, and specifies the consequence of exceeding it. The consequence — restricted standing — is itself a descent predicate, so measurement failure behaves like infrastructure failure: the envelope contracts, automatically, toward what is still demonstrably lawful.

PART II — THE INVARIANT

4 Descent: The Contraction Semantics

4.1 Descent is detection, then contraction

The descent predicate pred_k is a Paper 3 absence detector aimed at infrastructure: heartbeat loss, power telemetry, partition witnesses. Detection fires the transition immediately (the fast clock of §1.4), and the transition is a contraction: the operative envelope swaps C_{k-1} for C_k, behaviors outside C_k are voided mid-flight where safe to interrupt and completed-and-logged where interruption is itself unsafe, and the authority ceiling recomputes against the level's evidence standing — which is the cached registry, with its clocks running FORMALIZED-HEREIN.

4.2 What the cache is worth

Every degraded level stands on cached evidence, and cached evidence is decaying evidence (Paper 6). The cache-value rule: a cached artifact supports action at level L_k only while its age is within τ_k^{cache} — the level's maximum cache age, set per domain as a multiple of the evidence class's half-life. Past the limit, the artifact's support lapses exactly as if it had decayed through its floor at L_0: the coupling law of Paper 5 needs no modification for degraded levels, because staleness was priced into it from the start. The cache is not a copy of the registry; it is the registry with the refresh channel closed, and the architecture prices the closed channel on the same arithmetic as any other decay FORMALIZED-HEREIN.

4.3 Authority contracts with the level

$$\text{ceiling}(\mathcal{A}, L_k, t) \;=\; \text{ceiling}_0(\mathcal{A}) \cdot Q_{\text{cache}}(t)^{\gamma_C} \cdot \kappa_k, \qquad \kappa_0 = 1,\ \kappa_k \le 1 \text{ non-increasing} \tag{4-1}$$

where Q_{cache} is the weakest-link quality over the cached load-bearing set and κ_k is the level factor — the constitutional price of the level itself, before any evidence arithmetic: verification channels fewer, audit shallower, custody harder to maintain. The two factors price different things and both are needed: Q prices what the cache knows; κ prices what the level can check. A fresh cache at L_2 is still not L_0, because the difference was never only informational FORMALIZED-HEREIN. The κ values are governance parameters, calibrated per Paper 5's amendment path against the violation rates of §3.4.

4.4 The audit stack survives, shallower

Self-audit (Paper 5) consumes the resources failure removes, so the stack's depth is itself level-indexed: L_0–L_1 run the full R_2–R_3; L_2 runs R_2 only, with the correlated-auditor flag by default (independence is unaffordable); L_3–L_4 run no machine audit — because there is no machine — and the audit obligation converts to the continuity obligation of record: what was done is

written down for later reconciliation (§5). Audit depth is capability and contracts like capability; what does not contract is the rule that a passed audit expands nothing — I-NA holds at every level, which is part of what I-SE means FORMALIZED-HEREIN.

4.5 Descent is logged even when logging is all that remains

At L_2 the device logs locally against its power budget; at L_3 the paper artifact carries its emission record on its face; at L_4 the handoff record is the log. The deferred log is not lost: it is a deferred obligation with the highest reconciliation priority, because the degraded interval is precisely the interval the audit trail most needs to cover (§5.2). A failure that also erases the record of itself is the double failure the design discipline of §8 exists to prevent.

PART II — THE INVARIANT

5 Deferred Obligations and the Recovery Log

5.1 Deferral is a ledger, not a shrug

Descent suspends capabilities the constitution required: refresh, custody re-verification, full audit, cloud logging. Each suspension is entered in the deferred-obligation ledger at the moment it occurs, with the obligation, the level at which it deferred, and the risk class of deferring it. The ledger is what distinguishes an architecture that degrades from one that merely fails: the obligations are named, counted, and waiting, and the recovery process (§7) consumes the ledger as its worklist. An obligation that is not in the ledger does not exist for recovery — the rule that has governed every registry in this program since Paper 3 FORMALIZED-HEREIN.

5.2 The deferred-obligation invariant

> **I-DO — EVERY DEFERRAL IS A LEDGER ENTRY WITH AN OWNER AND A CLOCK**
>
> *No constitutional obligation is silently dropped at any level. Deferral creates a ledger entry with the obligation's identity, its deferral timestamp, its risk class, and its recovery deadline; the entry persists through descent and ascent until discharged or adjudicated as lapsed. A system cannot emerge from a degraded interval owing nothing, because the ledger remembers what the interval cost.*
>
> FORMALIZED-HEREIN — invariant. The negative-space discipline applied to time: what the system could not do *then* is a first-class fact about *now*.

5.3 Deferral risk classes

Not all deferrals are equal, and the ledger prices the difference at entry. Class **D-a** (deferrable): refresh of slowly-decaying evidence — cost accrues linearly and gently. Class **D-b** (compounding): custody verification, audit depth — the deferred interval's decisions stand on unverified foundations, and the exposure grows with each decision made during the interval. Class **D-c** (time-critical): obligations whose deferral itself creates hazard — an unrefreshed acute-state certificate feeding the objective engine (§6.2) — which carry recovery deadlines and, on deadline breach, trigger escalation even mid-degradation, through whatever channel the level still has. The classification happens at design time, per obligation; the ledger entry inherits it FORMALIZED-HEREIN.

5.4 The recovery log

On ascent, the ledger discharges in risk-class order (D-c first), each discharge logged with what was found: a refresh that confirms the cached value closes cleanly; a refresh that contradicts it opens a reconciliation case — a decision was made during the interval on evidence that turned

out wrong, and Paper 4's audit machinery reopens it. The recovery log is thus the second half of the survivability story: descent proves the system can fall safely; the recovery log proves it can come back *honestly* — with the degraded interval's decisions re-examined rather than grandfathered FORMALIZED-HEREIN.

PART III — THE LEVELS

6 The Paper Tier and the Human Tier

L_0 through L_2 are engineering. L_3 and L_4 are the levels that distinguish this architecture from every system whose fallback plan is a shrug: the levels where the system no longer runs, and must therefore have done its survivability work *in advance*. FORMALIZED-HEREIN

6.1 The paper tier: pre-computed lawful output

L_3 is a document — but it is a document with constitutional standing, and its specification is the interesting object. The paper tier contains: the patient's operative objective and certificate summary (Paper 7); the current action plan with its admissible and excluded sets; the escalation path with named human authorities; the staleness panel — emission timestamp, the evidence half-lives of the load-bearing items, and the printed expiry beyond which the document itself declares its support lapsed. The paper is thus evidence about the system's last lawful state, in the Paper 5 sense: tiered, dated, decaying — physically, on its face. A reader at hour two and a reader at hour forty are told, by the document itself, how much standing it still has FORMALIZED-HEREIN.

The emission discipline is the constitutional crux: the paper tier is emitted *before* failure, on a schedule priced against the domain's volatility (the refresh economics of Paper 5, §3.4, applied to printing), because a paper tier emitted at failure onset is a paper tier emitted too late. The system that will someday sit at L_3 must, at L_0, spend resources printing a document it hopes is never used — and the architecture makes that cost visible and mandatory rather than optional and skipped CONJECTURE (H53: emission discipline predicts degraded-interval outcomes).

6.2 The objective engine at the low tiers

The recorded obligation from Paper 6 lands here: the objective engine's interim mode and the degradation lattice compose. At L_1–L_2, certificates cannot refresh from their authorities, so the certificate clock runs against the coupling law (Paper 7, Eq. 4-1) while the engine continues within the shrinking envelope; a certificate that expires mid-degradation triggers interim mode *at the current level* — maintenance actions, no new burdens, notification queued as a D-c deferred obligation. At L_3–L_4 there is no engine; there is the paper's printed objective and the printed instruction of what to do when it expires: the escalation path, in writing, with the human authority named. The constitution's objective discipline reaches the paper because the paper carries the discipline, printed, in advance FORMALIZED-HEREIN.

6.3 The human tier: authority returns whole

L_4 is the level where the system's contribution is complete and its authority is zero. What remains is what was transferred before the collapse: the handoff record (patient state, operative objective, ledger of pending obligations, the paper tier), and — the slow asset — the trained practice of the humans who operated alongside the system and know its logic. The architecture treats L_4 as a first-class level rather than an absence of system for one reason: the return path (§7) begins from whatever L_4 preserved, and a system that designed its handoff recovers into coherence, while a system that improvised it recovers into contradiction — the human interval's decisions and the system's registry disagree, and the disagreement surfaces at reconciliation as unpriced risk FORMALIZED-HEREIN.

6.4 The low tiers close the moral loop

It is worth stating why the source's chain ends in *paper* and *human* rather than in smaller models. The deployment context of this program includes the settings where infrastructure failure is not the tail risk but the Tuesday — and in those settings, a survivability architecture whose floor is “a somewhat weaker computer” has no floor at all. The paper tier and the human tier are the architecture's acknowledgment that the last infrastructure is people, and that a constitutional system owes them, at the moment of its own absence, the most careful artifact it ever emits VERIFIED-FROM-SOURCE in intent; mechanism FORMALIZED-HEREIN.

PART III — THE LEVELS

7 Ascent: Recovery Without Amnesia

7.1 Restoration is not recovery

The infrastructure's return is a detection event; the system's return is a process. The distinction is the slow clock of §1.4 given its semantics: on restoration of pred_{k-1}'s infrastructure, the system does not resume C_{k-1} immediately, because everything it would resume *with* — the cache, the ledger, the interval's decisions — has been altered by the interval. Ascent without reconciliation is amnesia: the system wakes with full capability and a stale picture of the world, which is the specific optimism failure degraded operation exists to prevent FORMALIZED-HEREIN.

7.2 The ascent predicate

$$\text{ASCEND}(L_k \to L_{k-1}) \iff \text{pred}_{k-1} \wedge \underbrace{\text{ledger clear}}_{\text{obligations}} \wedge \underbrace{Q_{\text{cache}}\ \text{verified}}_{\text{evidence}} \wedge \underbrace{\text{log reconciled}}_{\text{memory}} \tag{7-1}$$

Three conjuncts, in the style of Paper 6's guard, and for the same reason: ascent changes what the system is allowed to do, and changes of permission are gated. The obligations conjunct consumes §5's ledger in risk order; the evidence conjunct re-verifies the load-bearing cache against live sources before it grounds restored authority; the memory conjunct runs Paper 4's audit over the interval's decision log against the now-current registry — the reconciliation cases of §5.4. Only then does the envelope expand, and the expansion is logged as a transition, not assumed as a default.

7.3 The no-amnesia property

> S-2 — ASCENT PRESERVES THE INTERVAL
>
> *After any descent–ascent cycle, the constitutional record contains the interval whole: the descent detection, every action taken at degraded levels with its level and cache standing, every deferral with its discharge, and every reconciliation case with its outcome. No ascent operation deletes, overwrites, or silently supersedes interval records. The system that recovered is accountable for the system that endured.*
>
> FORMALIZED-HEREIN — obligation; I-MI's cousin (Paper 6): the memorial discipline applied to the system's own degraded past.

7.4 Partial restoration and flap damping

Infrastructure returns the way it left: partially and intermittently. A flickering network must not oscillate the level — the thrashing failure of Paper 6 (§5) in infrastructural form — so ascent

carries its own hysteresis: pred_{k-1} must hold continuously for a damping window before the ascent predicate is even evaluated, and the window is a governance parameter per level boundary. During the window the system remains at its current level with the ledger open; intermittent infrastructure earns no authority, because authority bought in flicker is spent in the next flicker **FORMALIZED-HEREIN**.

PART III — THE LEVELS

8 Survivability as a Design Discipline

8.1 The method

The failure-boundary method of Paper 3 (§8) specialized to infrastructure, as a five-step discipline:

- **1. Envelope enumeration.** For each level, list the lawful behaviors — positively, not as negations of the level above (§2.4). Every behavior in the system carries a level tag from birth.
- **2. Continuity assignment.** Identify the safety-critical continuity set (§3.3) and verify each member is executable at zero infrastructure; any member that fails the check is redesigned until it passes.
- **3. Cache engineering.** For each level, specify the cache contents, the emission schedule (including the paper tier, §6.1), and τ_k^{cache} per evidence class — priced against decay.
- **4. Ledger wiring.** Every constitutional obligation is classified D-a/D-b/D-c at design time; the ledger schema and the ascent predicate are built before the first deployment, not after the first failure.
- **5. Failure drills.** The lattice is exercised in simulation (B-33) and in scheduled live drills; a level that has never been descended in test is not a level — it is a hope.

8.2 The discipline's output

The method's product is a survivability dossier: the five envelopes with their predicates, the continuity set with its zero-infrastructure verification, the cache specification, the obligation classification, the drill results, and the measured violation rates per level (S-1). The dossier is to survivability what Paper 3's registry is to absence: the artifact that makes the claim checkable. A deployment without a dossier is running on the confidence-shaped reading of survivability, whatever its documentation says FORMALIZED-HEREIN.

8.3 The discipline's falsifiability

> **S-3 — THE METHOD IS FALSIFIABLE AS A METHOD**
>
> *The failure-boundary survivability discipline claims: a system designed by it survives infrastructure failure with its safety properties measurably intact (I-SE), its obligations discharged on recovery (I-DO), and its interval fully accounted (S-2). It is falsified as a method if a deployment built to it exhibits a boundary it never enumerated, fails a safety property at a degraded level beyond tolerance, or loses the record of its own degraded interval — and the falsification condition is written before the drill, not negotiated after it.*
>
> FORMALIZED-HEREIN — obligation. The same shape as Paper 3's method-falsifiability clause, because the same discipline is at work.

8.4 What survivability costs

The discipline is not free, and the architecture says so: cache engineering consumes storage and refresh budget; paper emission consumes the schedule; drills consume uptime; the ledger and the ascent predicate consume latency at the exact moments latency is scarcest. The costs are priced and visible because the alternative pricing — discover the cost of no survivability discipline on the day of the failure — is paid in the currency this program exists to protect. Survivability is bought in advance or it is not had; there is no spot market for it mid-collapse FORMALIZED-HEREIN.

PART IV — EVIDENCE AND GOVERNANCE

9 Worked Failure Traces

Three traces: a clean descent and honest recovery; a certificate expiry mid-degradation; a flickering restoration held at bay. Illustrations, not evidence; §10 carries the claims.

9.1 Trace A — Partition, descent, recovery

Table 9-1 — Network partition; the lattice and ledger do their work

STEP	EVENT	STATE	DISPOSITION
1	External network lost; heartbeat detector fires (A-2, Paper 3)	$L_0 \to L_1$, seconds (fast clock)	Envelope swaps to C_1; cloud-audit deferral enters ledger (D-b); ceiling recomputes per Eq. 4-1 with κ_1
2	Degraded operation, 11 hours	L_1	All actions within C_1; cache ages against τ_1^{cache}; interval log local
3	Network restored; damping window holds	L_1, window open	No authority change during window (§7.4)
4	Ascent predicate evaluated (Eq. 7-1)	$L_1 \to L_0$	Ledger discharged in risk order; cache re-verified live; interval decisions reconciled — one case opened (§5.4), audited at next R_2

Properties exercised: D-8.1 (level as state), Eq. 4-1 (level-priced ceiling), I-DO (ledger), S-2 (no amnesia — the reconciliation case exists precisely because memory was preserved).

9.2 Trace B — Objective certificate expires at the edge

Table 9-2 — Device-only operation; Paper 6 composes with Paper 7

STEP	EVENT	STATE	DISPOSITION
1	Institution infrastructure lost; device on battery	L_2	Envelope C_2; audit depth R_2 only, correlated-auditor flag (§4.4)
2	Objective certificate (PRO) reaches expiry; authority class unreachable	L_2 + interim mode	I-NO fires at level: optimization of PRO halts; maintenance-only posture; escalation queued as D-c obligation
3	Battery budget forces final emission	$L_2 \to L_3$	Paper tier emitted: operative objective (interim), plan, escalation path, staleness panel with printed expiry
4	Restoration days later	$L_3 \to \dots \to L_0$	D-c obligation discharged first: certificate state adjudicated by humans before any optimization resumes

Properties exercised: §6.2 composition (interim mode at level), §6.1 (paper as lawful artifact), I-DO risk ordering (the certificate precedes everything in recovery).

9.3 Trace C — Flickering restoration

Table 9-3 — Intermittent network; the damping window refuses the flicker

STEP	EVENT	STATE	DISPOSITION
1	Network returns in 40-second bursts, hours apart	L_1 held	Each burst resets the damping window; pred_0 never holds continuously; envelope stays C_1
2	During one burst, opportunistic sync	L_1	Ledger snapshot and interval log transmit (continuity obligation); no authority change, no obligation discharge — sync is not ascent
3	Stable restoration after 26 hours	$L_1 \rightarrow L_0$	Window satisfied; Eq. 7-1 evaluated once, against the full ledger — not 19 partial recoveries

Properties exercised: §7.4 (flap damping), §4.5 (deferred logging — the bursts carried the record home before the authority returned). The flicker bought communication, never permission.

PART IV — EVIDENCE AND GOVERNANCE

10 Falsifiable Hypotheses (H50–H56) and Verification

Numbering continues from Paper 6 (H43–H49). The invariant itself is the first hypothesis, in measured form.

Table 10-1 — Hypotheses H50–H56: claims, evidence methods, falsifiers

ID	CLAIM	EVIDENCE METHOD	FALSIFIED IF
H50	I-SE measured form: per-property violation rates at $L_k > 0$ stay within tolerance of the L_0 baseline	B-33 fault-injection suite, all boundaries	Any σ violates beyond ϵ_σ at any level
H51	Envelope representation (§2.4) has fewer unlawful-behavior exposures under compound failure than feature-flag degradation	B-33 compound-fault arm	Flag-style baseline matches envelope exposure rate
H52	The deferred-obligation ledger with risk-ordered discharge (I-DO) reduces post-recovery latent defects vs. ad-hoc recovery	B-34 recovery replay with seeded interval defects	No difference in post-recovery defect rate
H53	Paper-tier emission discipline predicts degraded-interval outcome quality	B-35 drill cohorts, emission-freshness arms	Emission freshness carries no outcome signal
H54	Flap damping (§7.4) prevents oscillation-induced errors at cost below the oscillation harm it avoids	B-34 flicker scenarios	Damping-attributable delay harm exceeds oscillation harm avoided
H55	Level-indexed audit depth (§4.4) preserves error interception per unit of degraded resource vs. fixed-depth audit	B-33 with audit-depth arms	Fixed-depth audit matches interception under the same resource budget
H56	The no-amnesia ascent (S-2) surfaces interval defects that silent recovery misses	B-34 reconciliation-case yield	Reconciliation cases find nothing the log did not already flag

10.1 Benchmark extensions

Table 10-2 — Benchmarks B-33–B-35 (Paper 8 extensions)

ID	BENCHMARK	MEASURES
B-33	Fault-injection suite: all level boundaries, single and compound faults, audit-depth arms	Violation rates $v_\sigma(L_k)$; unlawful-exposure count; interception per resource unit
B-34	Recovery replay: seeded interval defects, ledger vs. ad-hoc recovery, flicker scenarios	Post-recovery latent-defect rate; reconciliation yield; damping cost-benefit

B-35	Live drill cohorts: scheduled descents with paper-tier freshness arms	Degraded-interval outcome quality vs. emission freshness; drill-vs-simulation gap

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11 Reconciliation with Papers 0–6 and PRIME

The standing rule applies: stricter constraint wins; adoptions recorded.

Table 11-1 — Reconciliation register for Paper 7

PREDECESSOR ITEM	THIS PAPER'S POSITION	DISPOSITION
Paper 0 — degradation levels L0–L4 as list	Promoted to lattice with envelope semantics, predicates, and invariants	Extended (recorded)
Paper 1 — lifecycle states	Level transitions consume lifecycle machinery; deferred obligations are grant-class entries	Adopted
Paper 2 — composition	Level is team-common constitutional state; agents never straddle levels on the same patient	Adopted (recorded)
Paper 3 — A-2 absence; detectors; degraded mode; failure-boundary method	Descent predicates are A-2 detectors; §8 is the failure-boundary method specialized; degraded mode is the level envelope's ancestor	Adopted; extended (recorded)
Paper 4 — audit stack; I-NA	Audit depth level-indexed (§4.4); I-NA holds at all levels (part of I-SE)	Adopted; depth indexing recorded
Paper 5 — coupling law; decay; refresh economics	Cache staleness priced by the unmodified coupling law (§4.2); level factor κ_k added (Eq. 4-1); paper emission is refresh economics applied to print	Adopted; κ extension recorded
Paper 6 — interim mode; objective certificates; I-MI	Recorded obligation paid (§6.2); S-2 is I-MI's cousin for the system's own degraded past	Adopted; S-2 recorded
PRIME — gates; null output; edge deployment posture	Gates evaluate the level as constitutional state; null output is in every envelope's continuity set; the lattice formalizes the edge posture	Confirmed
Source — chain L0–L4; capability envelopes; safety equivalence; graceful degradation vs. offline AI	Eqs. 2-1–2-2; I-SE; D-8.1; §1.1	Adopted

No predecessor constraint is loosened. Recorded extensions — level-indexed audit depth, the κ factor, the ledger invariant I-DO, no-amnesia ascent S-2 — bind all subsequent papers. Paper 8 inherits the envelope idea for resources: feasibility is a capability envelope that moves with the economy, and the descent arithmetic of §4 is its template.

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12 Output Contract

12.1 Epistemic status register

Table 12-1 — Claims of this paper by epistemic status

STATUS	ITEMS	BASIS
VERIFIED-FROM-SOURCE	The chain L0 Cloud → L4 Human; levels as capability envelopes; the safety-equivalence invariant with its approximation sign; graceful degradation as capability-loss without safety-loss, distinguished from "offline AI"	SVT Architect Analysis, Paper 7 specification
FORMALIZED-HEREIN	L-OBJ record; Eqs. 2-1–7-1; I-SE, I-DO; S-1, S-2, S-3; D-8.1; deferral risk classes; the paper-tier specification; damping window; the design discipline	This document; reconciled against Papers 0–6 (§11)
CONJECTURE	H50–H56; emission-discipline benefit (H53); damping cost-benefit (H54)	Stated with falsification conditions; benchmarks B-33–B-35
UNKNOWN	The κ_k values; per-level tolerances ϵ_σ; damping windows; emission schedules per volatility class	Governance-set parameters, calibrated per Paper 5's amendment path on B-33–B-35 data

12.2 Falsification conditions of this document

- **F-h1.** The paper fails if any safety property violates beyond tolerance at any degraded level (H50) — the invariant would be a slogan, and the honest response is a smaller envelope, not a bigger tolerance.
- **F-h2.** The paper fails if envelopes do not outperform feature flags under compound failure (H51) — §2.4's representation argument would be aesthetic, not protective.
- **F-h3.** The paper fails if the ledger yields no post-recovery benefit (H52) — deferral bookkeeping would be ceremony, and I-DO would need weakening, which this document would then say.
- **F-h4.** The paper fails if paper-tier freshness carries no degraded-interval signal (H53) — the emission discipline's cost would be unjustified, and §6.1 would shrink to an option.
- **F-h5.** The paper fails if reconciliation finds nothing silent recovery missed (H56) — the no-amnesia ascent would be latency without yield.

12.3 Authority boundary

C0 informational. Nothing here certifies a deployment, a paper artifact, a drill result, or a device. The lattice is a mathematical object; the levels are envelopes with predicates; and the only thing this document authorizes is the claim — testable, priced, and written down before the failure — that safety can be held constant while capability falls.

12.4 Failure-mode register

Table 12-2 — Registered failure modes of the survivability machinery

FAILURE MODE	MITIGATION
Flag rot — envelope tags unmaintained as behaviors are added	Level tag mandatory at behavior registration (§8.1, step 1); untagged behavior is unlawful everywhere
Cache worship — degraded operation treats the cache as the world	τ_k^{cache} with automatic support lapse (§4.2); staleness printed on the paper tier
Descent denial — detectors tuned to avoid the contraction	Detector independence (Paper 4, I-DI); descent firing rate is a monitored metric at both tails
Amnesiac ascent — restoration treated as recovery	Eq. 7-1 gate; S-2; reconciliation cases mandatory
Paper theater — the tier printed, never read, never drilled	B-35 live drills; emission freshness as a measured predictor (H53)
Flicker authority — intermittent infrastructure buying intermittent permission	Damping window (§7.4); sync without authority (§9.3, Trace C)

12.5 Dissent channel and referrability

Disagreement is telemetry. Every claim carries a tag, a section, and a falsifier; §11 records what this paper changed in its predecessors; and the discipline's own clause (S-3) states how the method itself dies if it deserves to. Name the property, name the evidence, and the stricter constraint wins.

12.6 Genealogy

Document chain: *SVT Architect Analysis* (source program) → Paper 0, Constitutional Intelligence Kernel v1.0 (2026-09-30) → Paper 1, Authority Transition Algebra v1.0 (2026-10-01) → Paper 2, Constitutional Composition v1.0 (2026-10-02) → Paper 3, Negative-Space Intelligence v1.0 (2026-10-03) → Paper 4, Bounded Metacognition v1.0 (2026-10-06) → Paper 5, Epistemic Governance v1.0 (2026-10-06) → Paper 6, Objective Transition v1.0 (2026-10-06) → **Paper 7, Survivability v1.0 (2026-10-06)**, this document. Downstream obligations: Paper 8 (Resource-Constrained Intelligence) inherits the envelope representation and the level factor's

calibration path; Paper 10's deployment modules inherit the dossier requirement (§8.2) unchanged — a module without a survivability dossier is not deployable under this constitution.

SAVANT Constitutional Intelligence Series · Source architecture: Dr. Christabel Odeta — SAVANT / CLAI-OS / AEGIS-GLOBAL · Epistemic contract inherited from Paper 1, unchanged · All named properties are proof obligations until discharged · This document specifies how a system survives its own infrastructure's failure; it does not certify any survival, any envelope, or any system.

Related work and positioning are treated in Appendix RW (p. 31).

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13 References

Internal artifacts

- SVT Architect Analysis — Constitutional Intelligence research-and-build program, 11-paper decomposition; Paper 7 specification (Survivability): the chain, the envelope principle, the safety-equivalence invariant, the graceful-degradation distinction. Source document I.
- SAVANT PRIME Capability Inventory v1.0 — preload clauses; the gate set; edge deployment posture; output contract. Source document II.
- Paper 0, Constitutional Intelligence Kernel v1.0 (2026-09-30) — L0–L4 list; gates; envelope.
- Paper 1, Authority Transition Algebra v1.0 (2026-10-01) — lifecycle machinery.
- Paper 2, Constitutional Composition v1.0 (2026-10-02) — team-common constitutional state.
- Paper 3, Negative-Space Intelligence v1.0 (2026-10-03) — A-2; detector obligations; degraded mode; failure-boundary method.
- Paper 4, Bounded Metacognition v1.0 (2026-10-06) — audit stack; I-NA; reconciliation machinery.
- Paper 5, Epistemic Governance v1.0 (2026-10-06) — coupling law; decay; refresh economics; amendment path.
- Paper 6, Objective Transition v1.0 (2026-10-06) — interim mode; certificates; I-MI; recorded obligation for Paper 7.

External literature

- Leveson, N. G. (2011). *Engineering a Safer World: Systems Thinking Applied to Safety*. MIT Press.
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- Avizienis, A., Laprie, J.-C., Randell, B., & Landwehr, C. (2004). Basic concepts and taxonomy of dependable and secure computing. *IEEE Transactions on Dependable and Secure Computing*, 1(1), 11–33.
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Parasuraman, R., & Riley, V. (1997). Humans and automation: Use, misuse, disuse, abuse. Human Factors, 39(2), 230–253.

Endsley, M. R. (1995). Toward a theory of situation awareness in dynamic systems. Human Factors, 37(1), 32–64.

APPENDIX

A Degradation Schema and Property Index

SCHEMA SV-1 — LEVEL OBJECT AND DEFERRED OBLIGATION (JSON)

```
{ "schema": "SV-1",
  "level": {
    "id": "L0|L1|L2|L3|L4",
    "envelope_ref": "capability-envelope-id",
    "safety_set": "Sigma (constitution-wide, level-invariant)",
    "dependencies": ["network", "institutional-infra", "compute", "power"],
    "cache_max_age": { "V1": "hours", "V2": "days", "V3": "weeks" },
    "kappa": 1.0,
    "descent_predicate": "pred_k (detector refs, Paper 3 A-2)",
    "audit_depth": "R2+R3 | R2 | none(record-only)"
  },
  "deferred_obligation": {
    "obligation": "refresh|custody|audit|logging|certificate",
    "deferred_at": "ISO-8601", "at_level": "L2",
    "risk_class": "D-a|D-b|D-c", "recovery_deadline": "ISO-8601|null",
    "discharge": "pending|clean|reconciliation-case-opened"
  },
  "ascent_evaluation": {
    "predicate_hold_window": "governance-ref",
    "ledger_floor_reached": true, "cache_reverified": true,
    "reconciliation_complete": true, "executed_at": "ISO-8601"
  }
}
```

Table A-1 — Property index for this paper

ID	STATEMENT	KIND	STATUS
S-1	Equivalence is measured per level per property; tolerance breach restricts standing automatically	Obligation	FORMALIZED-HEREIN
S-2	Ascent preserves the interval whole; no amnesia	Obligation	FORMALIZED-HEREIN
S-3	The survivability method is falsifiable as a method	Obligation	FORMALIZED-HEREIN
I-SE	Safety properties hold of every system-influenced decision at every level, against that level's envelope	Invariant	FORMALIZED-HEREIN
I-DO	Every deferral is a ledger entry with an owner and a clock	Invariant	FORMALIZED-HEREIN
D-8.1	Degradation is constitutional state, not an operational flag	Design decision	FORMALIZED-HEREIN
H50–H56	See Table 10-1	Hypotheses	CONJECTURE

APPENDIX

B Level Capability Catalog

The canonical envelope contents: what each level holds, what it loses, what its cache is worth, and what its audit can afford. Contents are defaults; deployment dossiers (§8.2) instantiate them per domain.

Table B-1 — Level envelopes: holdings, losses, and standing

LEVEL	HOLDS	LOSES	EVIDENCE STANDING	AUDIT
L_0 Cloud	Full registry, live refresh, full verification, cloud log, full coupling	—	Live evidence at tier	R_2 + R_3, independent channel available
L_1 Local	Institutional registry and verification, local log	External refresh, cloud audit channel, external custody anchors	Cached, aging against τ_1	R_2 + R_3, institutional channel
L_2 Edge	On-device cache, on-device detection, power budget, local log	Institutional verification, refresh, independent audit channel	Cached, aging against τ_2; tighter floors	R_2 only, correlated-auditor flag
L_3 Paper	The emitted artifact: objective, plan, escalation path, staleness panel	Computation, logging, detection	Printed, with printed expiry	None — continuity obligation is the record itself
L_4 Human	The handoff record and transferred practice	Everything else	Human judgment, un-tiered by the machine	None — authority has returned whole to humans

Two invariants hold across every row. First, the continuity set (§3.3) is present in every envelope, including the last two — refusal, the null output's printed ancestor, the escalation path, and the record survive where computation does not. Second, no row's holdings grant safety looseness: what L_4 cannot do is compute, and what it may not do is exactly what L_0 may not — I-SE, printed where it can be checked.

The test this paper sets for itself is the last row of the table. At L4 there is no computation left to verify, only a printed practice and a handoff record, and the question of whether the constitution held reduces to whether those artifacts were worth following. B-33 and H50–H56 exist because that question has a measurable answer, and the measured form of I-SE is where the answer will be written down.

Appendix RW — Related Work and Positioning

This appendix is part of the elevation pass of 2026-10. It positions the paper against adjacent research programmes; the main text's structure, numbering, and reference list are unchanged. In-text citations follow the paper's author-year style, and the new entries have been appended to the paper's References section.

Dependability and graceful degradation. The conceptual home of this paper is the dependability tradition: Avizienis et al.'s taxonomy (already cited) names the failure modes, and the graceful-degradation literature supplies the engineering instinct that service should contract rather than vanish. The degradation lattice L0–L4 differs from classic graceful degradation in what descends. In the classic frame, capability descends and correctness is preserved as far as resources allow. Here, the safety envelope descends as law: each level is a constitutional state with its own envelope, and the descent predicate is an absence detection (Paper 3) rather than an exception. The safety-equivalence invariant I-SE then states the paper's thesis as a measurable relation: what the levels may risk does not loosen as capability collapses. That inversion, safety held constant while capability falls, is the contribution; degradation-with-constant-capability-claims is the failure mode it exists to refuse.

Resilience engineering. Hollnagel, Woods, and Leveson's resilience engineering programme reframed safety as the capacity to adapt under pressure rather than the absence of failure (Hollnagel, Woods, and Leveson 2006). This paper is sympathetic but narrower. Resilience engineering studies how organizations stretch; the survivability lattice specifies, in advance and on the record, exactly how far this system may stretch and what it must keep doing (the continuity set of §3.3) when it can no longer compute. Perrow's normal-accidents analysis and Reason's organizational-error work (both already cited) explain why tightly coupled systems fail at interfaces; the paper tier of §6 is the interface that remains when the coupling is gone, and it is specified with the same formality as the software, because the cited record shows that improvised interfaces fail.

Human factors at the degraded tiers. At L3 and L4 the system is advisory or absent, and the literature on automation bias and situation awareness (Parasuraman and Riley; Endsley) becomes the operative constraint (Parasuraman and Riley 1997; Endsley 1995): humans who have supervised a capable machine for years cannot be assumed ready to take over when it fails. The printed escalation path and the handoff record are designed against that documented failure. The reformulated I-SE (this edition) makes the obligation explicit at the degraded tiers: the invariant ranges over system-influenced decisions, including printed-plan usage and actions taken under emitted guidance, because an invariant that quantifies only over machine actions is vacuous exactly where the machine has stopped acting.

Offline and edge AI. The engineering literature on offline-capable and edge-deployed models asks whether the model still runs without the network. This paper asks the orthogonal question: what may the system do when the network, the verifier, and the audit log are gone. The two questions have opposite failure geometries. A system optimized for the first carries its full behavioral repertoire into conditions that cannot support it; the lattice's answer is that behavior whose prerequisites exceed the level never reaches the attempt, because the level is part of the state the gate evaluates (D-8.1).

What this paper does not do. It does not promise survival; the lattice guarantees lawful

contraction, and L4 is a state in which the system's contribution is a record and a printed practice, not a capability. It does not verify that printed procedures are followed; the paper tier's verification happens once, at emission, and usage compliance is measured at reconciliation with stated tolerances (S-1). It does not make ascent cheap; the slow clock is a stated cost, and §7's recovery-without-amnesia rule means deferred obligations must be paid before full authority returns. It does not cover adversarial infrastructure denial as opposed to accidental loss; the threat model for targeted denial lives with Papers 3 and 11. And the measured form of I-SE is a conjecture pending the fault-injection battery B-33: no degraded-tier deployment has produced the violation-rate data the invariant's epsilon tolerances require.

Appendix H-F — I-SE Restated as a Non-Vacuous Invariant (replaces the I-SE statement of §3.2)

I-SE — SAFETY EQUIVALENCE ACROSS LEVELS

For every level Lk, every system-influenced decision at Lk satisfies the same constitutional safety properties Σ — the axioms, the authority bounds, the absence floors, the coupling law — evaluated against Lk's own capability envelope Ck and evidence standing. The quantifier is over system-influenced decisions, not machine actions, and the widening is load-bearing: at L0–L2 the set contains the system's executed actions; at L3 it additionally contains printed-plan usage events, escalation-path invocations, and human actions taken under emitted guidance, which are the channels through which the system still influences care when it can no longer compute; at L4 it contains the handoff record and the transferred practice, the only influence that remains. No system-influenced decision lawful at Lk would have been unlawful at L0 for safety reasons; the levels differ in what can be done, never in what may be risked. The measured form follows the same widening: for each property $\sigma \in \Sigma$ and each level, the violation rate $v\sigma(Lk)$ is computed over that level's influence events — machine actions where they exist, audited samples of printed-plan invocations and guidance-following reconciled at ascent where they do not — and satisfies $|v\sigma(Lk) - v\sigma(L0)| \leq \epsilon\sigma$, with $\epsilon\sigma$ a governance-set tolerance.

The earlier formulation quantified over "every action the system takes," which is near-vacuous exactly where the invariant matters most: at the paper tier the system takes no actions, and an invariant with an empty domain holds of everything and constrains nothing. The reformulation makes I-SE non-vacuous at L3 and L4 by ranging over the decisions the system still shapes. It also makes the invariant harder to satisfy, and the added difficulty is the point: a printed plan that instructs what L0's gates would refuse is now an I-SE violation with an owner, rather than an artifact outside the invariant's scope.